6 b output.sci (C:\Users\LENOVO\Documents\6 b output.sci) - SciNotes

*6 b output.sci

```
end

// Non-Causal System: y[n] = 0.5x[n] + 0.5x[n+1]
for k = 1:N
if k < N then
y_noncausal(k) = 0.5 * x_n(k) + 0.5 * x_n(k+1);
else
y_noncausal(k) = 0.5 * x_n(k); // No x[n+1] for k=N-1
end
end
// Plot the input sequence
clf; // Clear the current figure
subplot(3,1,1);
plot2d(n, x_n);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Input Sequence: Unit Step Sequence');
// Plot the output of the causal system
subplot(3,1,2);
plot2d(n, y_causal);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Output Sequence: Causal System');
// Plot the output of the non-causal system
subplot(3,1,3);
plot2d(n, y_noncausal);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Output Sequence: Non-Causal System');

```

Line 46, Column 0.

Graphic window number 0

File Tools Edit ?

